

Masked k-NN Imputation with `missknn`

Distance masking, per-column tuning, and donor-pool capping for mixed tabular data.

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Mixed tabular data with missing values, $X = (x_{ij}) \in \mathbb{R}^{n \times p}$, is usually imputed with an iterative random-forest engine such as `missForest`, which needs no distributional assumptions but scales poorly with n and can fail outright when a rare categorical level is entirely absent from a training fold. `missknn` targets the same mixed numeric and categorical setting with a k -nearest-neighbor engine instead, built so that distances only ever compare jointly observed variables, and so that neighborhood size and estimator are chosen per column by internal holdout evaluation rather than fixed in advance.¹

Masked distance

Let $R_{ij} = \mathbb{1}[x_{ij} \text{ is observed}]$ and $O_i = \{j : R_{ij} = 1\}$. For a receiver row i with a missing target t , and a donor row ℓ with $R_{\ell t} = 1$, the shared observed set excluding the target itself is

$$S_{i\ell t} = O_i \cap O_\ell \setminus \{t\}.$$

Pairs with $S_{i\ell t} = \emptyset$ share nothing comparable and are discarded. Numeric columns are standardized from observed values only, $z_{ij} = (x_{ij} - \mu_j)/\sigma_j$, and each shared variable contributes

¹`missknn` package documentation, CRAN: <https://CRAN.R-project.org/package=missknn>.

$$\delta_j(i, \ell) = \begin{cases} (z_{ij} - z_{\ell j})^2 & j \text{ numeric} \\ \mathbb{1}[x_{ij} \neq x_{\ell j}] & j \text{ categorical.} \end{cases}$$

The masked distance is a weighted rate over the shared observed mass, not a sum discounted by how much is missing:

$$d_t(i, \ell) = \sqrt{\frac{\sum_{j \in S_{i\ell t}} w_j \delta_j(i, \ell)}{\sum_{j \in S_{i\ell t}} w_j}},$$

with weights $w_j \geq 0$, uniform by default. Normalizing by $\sum_{j \in S_{i\ell t}} w_j$ rather than a fixed $p - 1$ means a donor pair with less overlap is not automatically penalized: distance measures how different two rows are on what is actually comparable between them, the same role partial matching plays in Gower's coefficient.²

Neighbor aggregation

Given $d_t(i, \cdot)$, let $N_k(i, t)$ be the k donors with smallest finite distance. For a numeric target, the default estimator is an inverse-distance-weighted mean,

$$\hat{x}_{it} = \frac{\sum_{\ell \in N_k(i, t)} \alpha_{i\ell t} x_{\ell t}}{\sum_{\ell \in N_k(i, t)} \alpha_{i\ell t}}, \quad \alpha_{i\ell t} = \frac{1}{d_t(i, \ell) + \varepsilon},$$

and for a categorical target a weighted-majority vote, $\hat{x}_{it} = \arg \max_v \sum_{\ell \in N_k(i, t)} \alpha_{i\ell t} \mathbb{1}[x_{\ell t} = v]$. Under multiple imputation ($m > 1$), each completed dataset draws a donor from $N_k(i, t)$ with probability proportional to $\alpha_{i\ell t}$ instead of collapsing to the point estimate, so imputation uncertainty is propagated rather than repeated m times.

Local regression

A neighbor average ignores how the target co-varies with other predictors. `missknn` also fits a local ridge regression over a wider neighborhood $N_{k'}(i, t)$, $k' > k$:

$$\hat{\beta} = \arg \min_{\beta} \sum_{\ell \in N_{k'}(i, t)} (x_{\ell t} - \beta^\top \mathbf{z}_\ell)^2 + \lambda \|\beta_{-0}\|_2^2,$$

where $\mathbf{z}_\ell$ stacks the intercept with observed numeric predictors and the ridge penalty λ leaves the intercept unregularized. Neighbors are weighted uniformly here rather than by inverse

²Gower, J. C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, 27(4), 857-871.

distance: a near-duplicate donor is a stabilizing dominant vote for a mean, but the same near-duplicate makes the regression's normal equations nearly singular, so the weighting scheme that helps the mean estimator would actively hurt the regression one.

Per-column tuning

Neither k , k' , nor the choice between the mean and the regression estimator is fixed globally. For each column, `missknn` holds out roughly 25 percent of that column's observed entries, capped at 100 rows, imputes them from the remaining donors under each candidate k , and scores holdout error. An improvement is only accepted once it clears a fixed relative margin over the running best, 8 percent for neighborhood size and 20 percent for switching to regression, because with finite holdout data a smaller k or a regression fit can look marginally better than the incumbent by chance alone, and the larger margin for regression reflects that it is being compared as the better of two noisy candidates rather than a single alternative.

The same holdout scoring also evaluates the plain global mean or mode of the column. If nothing clears the margin against this baseline, the column is filled directly from it and neighbor search is skipped entirely, both because a k -point average carries needless variance when a column has no local signal to exploit, and because the global fill costs $O(1)$ per receiver rather than a full neighbor search. For columns that do need neighbor search, an unbounded search across all donors is $O(n_{\text{recv}} \times n_{\text{don}})$. When the donor pool exceeds a cap (2000 by default), a random subsample of that size is drawn once per column and reused for every receiver, keeping cost linear in the number of receivers rather than quadratic in n .

```
Require: X, R, weights w, candidate grid K, donor cap C
D <- donors of column t; M <- receivers of column t
if |D| = 0: fill M with a constant default; return
(H, T) <- random holdout/train split of D
e* <- holdout error of global mean/mode; k* <- null; est* <- global
for k in K:
  e_k <- holdout error of the weighted mean at neighborhood k
  if e_k < e*(1 - 0.08): e* <- e_k; k* <- k; est* <- mean
k'_narrow <- max(4 k*, 30); k'_wide <- min(n_train, 300)
e_reg <- best holdout error of the ridge regression at k'_narrow, k'_wide
if e_reg < e*(1 - 0.20): est* <- regression
if est* = global: fill M from the global mean/mode of D; return
D' <- random subsample of D of size min(|D|, C)
for i in M: rank D' by masked distance; fill x_it via the mean or the regression estimator
```

This procedure runs once per column on the first pass, and once more per completed dataset when $m > 1$, with stochastic donor draws in place of the point estimate.

Example

The `missknn` package exposes a single constructor with a `complete()` accessor.

```
library(missknn)

set.seed(1)
x <- data.frame(
  a = c(rnorm(20, 10, 2)),
  b = c(rnorm(20, 5, 1)),
  g = factor(sample(c("low", "mid", "high"), 20, replace = TRUE))
)
x$a[sample(20, 4)] <- NA
x$b[sample(20, 3)] <- NA
x$g[sample(20, 2)] <- NA

imp <- missknn(x, k = 5, m = 1, numeric_estimator = "regression", seed = 1)
imp
```

```
<missknn>
 observations: 20
 variables: 3
  k: 5
completed datasets: 1
 scale: TRUE
add_indicator: FALSE
```

The completed data replaces every missing entry with its estimate.

```
out <- as.data.frame(complete(imp))
out$a <- round(out$a, 2)
out$b <- round(out$b, 2)
head(out, 8)
```

	a	b	g
1	8.75	5.92	mid
2	10.37	5.78	low
3	8.33	5.07	high
4	13.19	3.01	low
5	10.66	5.62	low
6	8.36	4.94	low

```
7 11.20 5.68 low
8 11.48 3.53 high
```

`summary()` reports how much was missing per column and the settings used for the fit.

```
str(summary(imp))
```

```
List of 8
 $ n          : int 20
 $ p          : int 3
 $ k          : int 5
 $ m          : int 1
 $ scale      : logi TRUE
 $ add_indicator : logi FALSE
 $ missing_by_variable: Named int [1:3] 4 3 2
 ..- attr(*, "names")= chr [1:3] "a" "b" "g"
 $ missing_total : int 9
 - attr(*, "class")= chr "summary.missknn"
```

Practical notes

The donor cap and the holdout-driven global fallback are what keep runtime near-linear in n rather than the amount of available parallelism: `missknn` has no user-facing thread-count argument, and the compute-heavy steps are parallelized internally per receiver regardless of core count. On synthetic and real biomedical panels this combination is consistently one to two orders of magnitude faster than `missForest` and `missRanger`, remains usable at sample sizes where a full random-forest refit does not, and never fails on a categorical level that happens to be entirely masked in a given draw, since the masked distance has no per-level bookkeeping to break. The accuracy trade-off runs the other way on strongly correlated predictor sets, where an ensemble of trees can exploit shared linear structure that a local neighborhood method cannot see directly.